

Bank Loan Approval Prediction: Project Documentation

1. Executive Summary

1.1 Objective

The primary objective of this project is to build an automated machine learning credit risk evaluation system that predicts loan approval eligibility based on applicant demographics, financial background, loan parameters, and historical credit behavior. Automating this decision pipeline increases underwriting efficiency while maintaining credit quality.

1.2 Baseline Performance Overview

The baseline model—a Random Forest Classifier with 100 estimators—achieved an overall accuracy of 78.05% on the test set (123 samples). However, granular performance metrics reveal significant asymmetry as shown in the table below:

Metric	Approved Loans (Class 1)	Rejected / High-Risk (Class 0)
Recall	97%	42%
Precision	76%	90%
F1-Score	0.85	0.57
Support	80 samples	43 samples

Critical Risk Finding: The model failed to identify 58.1% (25 of 43) of high-risk / rejected applicants, misclassifying them as approved loans. In retail banking, this represents substantial credit risk and potential default exposure.

2. Dataset Architecture & Feature Dictionary

2.1 Dataset Overview

- **Total Records:** 614 loan applications
- **Total Input Features:** 12 attributes
- **Target Variable:** Loan_Status (Binary: Approved / Rejected)
- **Total Missing Data Points:** 149 missing values across multiple columns

2.2 Feature Specifications

Feature Name	Description	Type / Scale
Loan_ID	Unique identifier for each loan application	String / Nominal ID
Gender	Applicant gender	Male / Female
Married	Marital status	Yes / No
Dependents	Number of dependent family members	0, 1, 2, 3+
Education	Applicant educational attainment	Graduate / Not Graduate
Self_Employed	Employment category	Yes / No
ApplicantIncome	Monthly gross income of primary applicant	Continuous
CoapplicantIncome	Monthly gross income of co-applicant	Continuous
LoanAmount	Principal loan amount requested (in thousands)	Continuous

Feature Name	Description	Type / Scale
Loan_Amount_Term	Repayment schedule term in months	Discrete (e.g., 360)
Credit_History	Guideline compliance indicator	1.0 (Met), 0.0 (Failed)
Property_Area	Property location classification	Urban, Semiurban, Rural

3. Baseline Pipeline & Technical Audit

3.1 Implemented Workflow

1. **Ingestion:** Loaded dataset into Pandas DataFrame.
2. **Global Encoding:** Applied sklearn LabelEncoder across all object-type columns prior to partitioning.
3. **Partitioning:** Executed 80/20 train-test split with fixed random seed (random_state=42).
4. **Training:** Fitted RandomForestClassifier(n_estimators=100, random_state=42).
5. **Evaluation:** Measured accuracy (78.05%), classification report, and confusion matrix.

3.2 Key Technical Issues & Vulnerabilities

- **Feature Leakage & Spurious Correlation:** Loan_ID was left in the feature matrix X during model training. Identifiers have no predictive relationship to creditworthiness and risk fitting to index artifacts.
- **Preprocessing Data Leakage:** LabelEncoder was fit on the full dataset before splitting into train and test sets, violating data isolation best practices.
- **Arbitrary Ordinality:** Nominal categorical features (e.g., Property_Area, Gender) were encoded as sequential integers (0, 1, 2), imposing artificial mathematical ordering.

- **Missing Value Handling:** The 149 missing values were not explicitly imputed with domain-appropriate strategies (e.g., median for skewed incomes, mode for categorical variables).

4. Evaluation & Asymmetric Risk Impact

4.1 Detailed Performance Metrics

Metric Type	Value / Score
Accuracy	78.05%
Macro Avg Precision	0.83
Macro Avg Recall	0.70
Macro Avg F1-Score	0.71
Weighted Avg Precision	0.81
Weighted Avg Recall	0.78
Weighted Avg F1-Score	0.75

Confusion Matrix Summary:

- True Negative (Rejected correctly): 18
- False Positive (Rejected predicted as Approved): 25
- False Negative (Approved predicted as Rejected): 2
- True Positive (Approved correctly): 78

4.2 Financial Risk Breakdown in Banking

- **Type I Error (False Positive / False Approval):** High-risk applicants approved for loans. Results in capital loss, loan default, write-offs, and costly recovery procedures. (**Severity: Critical**)

- **Type II Error (False Negative / False Rejection):** Creditworthy applicants rejected. Results in forgone interest revenue and potential applicant friction. (Severity: Low to Moderate)

Conclusion: A 42% recall on high-risk applicants exposes the institution to unacceptable default rates.

5. Engineering Roadmap & Production Remediation

5.1 Preprocessing & Feature Engineering

- **Pipeline Isolation:** Build a modular Scikit-Learn ColumnTransformer and Pipeline.
- **Feature Removal:** Drop Loan_ID.
- **Imputation:** Median imputation for continuous variables (ApplicantIncome, CoapplicantIncome, LoanAmount); Mode imputation for categorical features and Credit_History.
- **Feature Transformations:** Apply log/power transforms to skewed income and loan amounts, followed by RobustScaler / StandardScaler.
- **Categorical Encoding:** One-Hot Encoding with drop='first' for nominal categories.
- **Domain Features:**
 - $\text{Total_Income} = \text{ApplicantIncome} + \text{CoapplicantIncome}$
 - $\text{Debt_To_Income_Ratio} = \text{LoanAmount} / \text{Total_Income}$
 - $\text{EMI_To_Income_Ratio} = \text{Estimated_Monthly_Payment} / \text{Total_Income}$
 - $\text{Income_Per_Dependent} = \text{Total_Income} / (\text{Dependents} + 1)$

5.2 Algorithmic Optimization & Threshold Calibration

- **Class Weighting:** Incorporate class_weight='balanced' in tree ensembles or apply SMOTE resampling.
- **Model Comparison:** Benchmark against XGBoost, LightGBM, CatBoost, and L1/L2 Regularized Logistic Regression.
- **Threshold Tuning:** Shift probability decision thresholds to prioritize higher recall on Class 0 (targeting $\geq 80\%$ recall on rejections).

5.3 Governance, Explainability & Compliance

- **SHAP Integration:** Use SHapley Additive exPlanations to compute global feature importance and applicant-specific local attributions.
- **Regulatory Compliance:** Comply with Fair Lending guidelines (ECOA/FCRA) by auditing protected demographic features and generating automated reason codes for Adverse Action Notices.